

File-by-File Project Audit

Exhaustive Inventory with Necessity Classification

Every source file, configuration, asset, and data artifact

Legend

REQUIRED — app will not run without it

OPTIONAL — useful but non-essential

DEAD — unused; safe to delete

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1 Methodology

Every file in this report has been classified as one of three statuses:

- **REQUIRED** — the live application, test suite, or development environment will fail or become non-functional if this file is removed. Examples: Django settings, URL routing, model definitions, React entry points.
- **OPTIONAL** — removing this file reduces capability (loses tests, seed data, a debug command, a migration script) but the running application continues to work. Example: data loaders, optional admin commands, test fixtures.
- **DEAD** — not referenced by any live import or runtime path. Safe to delete. Typically: development-time debug scripts, stale output logs, abandoned prototypes.

1.1 Excluded directories

Large auto-generated or irrelevant directories are not enumerated file-by-file. They are listed below as single entries.

.git/	n/a
Purpose: Git version control metadata — object store, refs, hooks, and configuration for every commit in the repository history. <i>If deleted:</i> Destroys the repository history; project becomes a plain directory with no version control.	
frontend/node_modules/	REQUIRED for dev
Purpose: NPM-installed frontend dependencies (<code>react</code> , <code>react-dom</code> , <code>react-router</code> , <code>recharts</code> , <code>react-simple-maps</code> , <code>jspdf</code> , <code>tailwindcss</code> , etc.). Reinstallable from <code>package-lock.json</code> . <i>If deleted:</i> Frontend dev server cannot start; <code>npm install</code> recreates it.	
backend/node_modules/	DEAD
Purpose: Stray NPM dependencies installed at the backend root for a legacy Node utility (likely <code>generate_comparison_pdf.js</code>). The live backend is Python only; none of these packages are imported by Django. <i>If deleted:</i> Nothing breaks. Safe to delete entirely, saves hundreds of MB.	
backend/.venv/	REQUIRED for dev
Purpose: Project-local Python virtual environment with installed pip dependencies (Django, DRF, sentence-transformers, setfit, keybert, gliner, rapidfuzz, etc.). <i>If deleted:</i> Backend cannot run until dependencies are reinstalled with <code>pip install -r requirements.txt</code> .	
backend/search/models/intent_model/	REQUIRED
Purpose: Trained SetFit intent classifier (<code>model.safetensors</code> , <code>model_head.pkl</code> , <code>config_setfit.json</code> , tokenizer files, ... 129 MB total). Loaded at startup by <code>nlu_engine.ModernNLUEngine</code> . <i>If deleted:</i> NLU falls back to the regex intent detector; functional but misses subtleties on ambiguous phrasing.	
frontend/build/	OPTIONAL
Purpose: Output of <code>npm run build</code> — minified production bundle (<code>static/js/*.js</code> , <code>static/css/*.css</code> , HTML shell). Regenerable anytime. <i>If deleted:</i> Production static bundle is gone until <code>npm run build</code> is re-run; dev server (<code>npm start</code>) still works.	
__pycache__/ (every Python package)	n/a
Purpose: Python bytecode cache automatically regenerated on import. Every Django app, service, and script directory has one. <i>If deleted:</i> Deleted and regenerated freely; no impact.	
migrations/__pycache__/	n/a
Purpose: Same as above but for migration files. <i>If deleted:</i> Same; regenerated.	
.pytest_cache/	n/a
Purpose: Pytest's last-run cache. <i>If deleted:</i> Regenerated on the next test run.	
.claude/	n/a
Purpose: Claude Code session state for this repository. Editor tool, not project code. <i>If deleted:</i> Claude Code loses session history for this repo; project is unaffected.	

1.2 File counts

After exclusions above, the audit covers approximately:

- **Project root:** ~10 files
- **Backend Django apps** (9 apps): ~120 files
- **Backend search engine:** 8 Python source files + 1 trained model bundle
- **Backend root:** ~10 files (mostly config)
- **Frontend source tree:** ~75 React components, hooks, services, and styles
- **Other directories** (scripts, Resources, docs, .github): ~15 files

2 Backend — Search Engine (backend/search/)

The search app is the ML-powered core of the product: it exposes a single DRF viewset (`/api/search/`) that takes a natural-language query (e.g. “*buy cheap sugar from china under \$500*”) and returns a ranked list of supplier/buyer profiles aggregated from the `trade_data.Transaction` table. The pipeline is NLU → subcategory resolution → ORM aggregation → composite ranking. The 8 source files below are the live files in this app after the recent dead-code cleanup removed `nlp.py`, `vector_store.py`, `candidate_retrieval.py`, `ranking.py`, `ranking_ltr.py`, `train_ltr.py`, `ltr_dataset_builder.py`, `build_search_index.py`, `train_intent.py`, and `query_parser.py`.

2.1 Package roots

`backend/search/__init__.py`

REQUIRED

Purpose: Empty file marking `search` as a Python package so Django can discover the app.

If deleted: Django raises `ModuleNotFoundError` on startup; all search endpoints fail.

`backend/search/apps.py`

REQUIRED

Purpose: Django `AppConfig` registering the app under the name `search`. Referenced in `settings.INSTALLED_APPS`.

If deleted: Django refuses to load the app; `/api/search/` routes disappear.

`backend/search/models.py`

REQUIRED

Purpose: Essentially empty — just `from django.db import models`. Django requires each app to have a `models.py` file even if no models are defined. The search app intentionally stores no state of its own; all state lives in `trade_data.Transaction`.

If deleted: Django management commands (`makemigrations`, `check`) warn that the app is misconfigured.

`backend/search/urls.py`

REQUIRED

Purpose: Registers the DRF `DefaultRouter` for `SearchViewSet` at the empty prefix and explicitly wires `supplier-detail/` to avoid router ambiguity.

If deleted: All search API routes 404. The frontend’s entire search page becomes non-functional.

`backend/search/views.py`

REQUIRED

Purpose: `SearchViewSet` — the live DRF entry point. Provides `list` (main search), `autocomplete` (product suggestions), `supplier_detail` (company deep-dive), and `supplier_compare`. Also holds the module-level `global_search_service` singleton so ML models load once at Django boot.

If deleted: Every search-related HTTP endpoint returns 404 or 500.

2.2 Services layer

`backend/search/services/__init__.py`

REQUIRED

Purpose: Package marker for the `services` sub-package.

If deleted: Imports from `search.services.*` fail.

`backend/search/services/search_service.py`

REQUIRED

Purpose: The main orchestrator — class `SearchService`. Contains `execute_search()` (the 4-stage pipeline: NLU, subcategory resolution, ORM aggregation, composite ranking), `_resolve_subcategories()` (startswith/icontains/pg_trgm passes with the disambiguation short-circuit), `_orm_search()` (aggregation + ranking with Redis cache), and `autocomplete_products()` (single-query annotated autocomplete).

If deleted: The view layer cannot execute any search. The entire search feature is down.

`backend/search/services/nlu_engine.py`

REQUIRED

Purpose: `ModernNLUEngine` — the natural language parser. SetFit intent classifier (with regex pattern rules for semantic-reversal phrases like “looking for sellers of X”), KeyBERT product-keyword extraction with 40 stopwords, GLiNER entity extraction (loaded but mostly unused), RapidFuzz country resolution with typo tolerance, OpenRouter LLM unified parser (optional), and `_build_price_filter_regex` as the deterministic price fallback. Also exposes `extract_product_keyword` — a pure-regex helper.

If deleted: `search_service.execute_search` crashes on the first line of its NLU stage; every query fails.

`backend/search/services/aggregation.py`

REQUIRED

Purpose: `SupplierAggregator` — pure Django-ORM aggregation over the `Transaction` table. Provides `get_suppliers_for_subcategories()` (main list), `get_supplier_details()` and `get_buyer_details()` (deep-dive stats + sparklines + intelligence metrics), and `get_supplier_comparison()` (side-by-side). Applies the intent/scope → trade_type mapping and the post-aggregation price HAVING filter.

If deleted: `search_service` cannot produce supplier lists or detail pages; the entire search + company pages become non-functional.

2.3 Management commands

`backend/search/management/__init__.py`

REQUIRED

Purpose: Marks management as a package so Django discovers the nested `commands` directory.

If deleted: Django cannot load custom commands from this app.

`backend/search/management/commands/__init__.py`

REQUIRED

Purpose: Marks the `commands` directory.

If deleted: `python manage.py sync_search` fails with `Unknown command`.

`backend/search/management/commands/sync_search.py`

OPTIONAL

Purpose: Indexes every `Transaction` row into OpenSearch as a hybrid BM25 + kNN index (`all-MiniLM-L6-v2` embeddings, 384-d vectors). Only relevant when the OpenSearch path is enabled — currently the live search path has `if False` hardcoded, so this command is a dormant utility.

If deleted: No effect on the running app. If you ever re-enable OpenSearch, the feature can't be initialized.

2.4 Tests

`backend/search/tests/__init__.py`

OPTIONAL

Purpose: Marks the `tests` package for pytest/Django test discovery.

If deleted: Tests stop being auto-discovered.

`backend/search/tests/test_scope_consistency.py`

OPTIONAL

Purpose: Single unit test that mocks `Transaction.objects` and verifies the aggregator passes the `product_item_filter` correctly through the Django ORM chain. Prevents a regression where the detail page and the list page would disagree on shipment counts.

If deleted: No impact at runtime. You lose one regression test.

2.5 Migrations

`backend/search/migrations/` (directory)

REQUIRED

Purpose: Standard Django migrations directory. The search app defines no models of its own, so this contains only an `__init__.py` marker.

If deleted: `makemigrations` complains that the app has no migrations module.

2.6 Trained ML models

`backend/search/models/intent_model/` (directory)

REQUIRED

Purpose: The trained SetFit BUY/SELL intent classifier (BAAI/bge-small-en-v1.5 backbone + sklearn logistic head). Contains the SentenceTransformer body (`model.safetensors`, `config.json`, `tokenizer.json`, `vocab.txt`, `special_tokens_map.json`, `sentence_bert_config.json`, `modules.json`, `config_sentence_transformers.json`, `1_Pooling/`, `2_Normalize/`) and the head (`model_head.pkl`, `config_setfit.json`) — 129 MB total.

If deleted: NLU falls back to the regex intent detector. Basic BUY/SELL queries still work, but the model's generalization to unusual phrasings is lost.

3 Project Root

This section covers every file at the top level of the repository (not inside any subdirectory).

3.1 Configuration and Docker

.gitignore

REQUIRED

Purpose: Repository-wide Git ignore rules — excludes `.venv/`, `node_modules/`, `__pycache__/`, compiled assets, credentials, etc.

If deleted: Git starts tracking junk (venvs, node modules, build artifacts), polluting commits.

.dockerignore

OPTIONAL

Purpose: Excludes files from the Docker build context when building an image from the project root.

If deleted: Only matters if you ship a root-level Docker image. Larger build contexts otherwise.

docker-compose.yml

OPTIONAL

Purpose: Spins up OpenSearch + OpenSearch Dashboards for local dev. The live search pipeline has the OpenSearch path disabled (if `False`), so this is a dormant utility. Required only if you plan to re-enable OpenSearch as the BM25 fast path.

If deleted: Local OpenSearch stack can no longer be brought up via `docker compose`. The running app is unaffected.

3.2 Documentation

README.md

OPTIONAL

Purpose: Main repository README — project overview and quick-start instructions. 118 KB; partially outdated.

If deleted: New contributors lose their landing-page orientation.

SUGGESTED_IMPROVEMENTS.md

OPTIONAL

Purpose: Suggested-improvements document (filename has a typo: “IMRP” instead of “IMPR”).

If deleted: No runtime impact.

SRS Main.docx

OPTIONAL

Purpose: Word version of the Software Requirements Specification. Academic deliverable for the FYP.

If deleted: Lose the editable version of the SRS.

SRS Main.pdf

OPTIONAL

Purpose: PDF export of the SRS.

If deleted: Lose the read-only version of the SRS.

3.3 Setup scripts

setup_linux.sh

REQUIRED

Purpose: Linux setup script: creates the venv, installs dependencies, initializes the Postgres database, runs migrations, seeds data.

If deleted: Fresh-environment setup on Linux no longer works end-to-end.

setup_mac.sh

OPTIONAL

Purpose: macOS equivalent.

If deleted: macOS contributors must set up manually.

setup_windows.ps1

OPTIONAL

Purpose: Windows PowerShell equivalent.

If deleted: Windows contributors must set up manually.

3.4 Data inputs at root

companies.xlsx

OPTIONAL

Purpose: Seed company list (21 KB) — imported by `load_data.py`.

If deleted: Fresh DB seeding of the companies table fails.

keycontacts.xlsx Purpose: Seed key-contacts list imported by <code>load_data.py</code> . <i>If deleted:</i> Fresh DB seeding of the key contacts fails.	OPTIONAL
import_data_1year.xlsx Purpose: Raw 1-year import transactions (334 KB), used as source data for seeding the Transaction table. <i>If deleted:</i> Cannot re-seed transactions from scratch.	OPTIONAL
export_data.xlsx Purpose: Raw export transactions (665 KB). The current DB is import-only, so this is prepared for a future ingestion path that is not yet wired up. <i>If deleted:</i> Lose a staging source for future export ingestion.	OPTIONAL

3.5 Python utility scripts at root

4 Backend — Top-Level Files (backend/*)

Everything under `backend/` that is not inside a Django app directory. After the April 2026 dev-cruft cleanup, this section now contains only live configuration files, the GNN graph data files used by `trade_ledger`, the Puppeteer PDF-export script, and a few optional seed/recovery helpers. The ~60 debug scripts and ~60 stale output logs that previously lived here have all been removed.

4.1 Live Django machinery

`backend/manage.py`

REQUIRED

Purpose: Django management entry point. Also contains the env-var preamble that suppresses TensorFlow / transformers / matplotlib warnings at startup.

If deleted: Project cannot run (`python manage.py runserver` fails).

`backend/.env`

REQUIRED

Purpose: Environment variables for Django (`SECRET_KEY`, optionally `OPENROUTER_API_KEY`, DB overrides, etc.). Loaded by `python-dotenv` in `settings.py`.

If deleted: `settings.SECRET_KEY` is empty; Django refuses to boot.

`backend/.gitignore`

REQUIRED

Purpose: Backend-specific Git ignore rules (`__pycache__`/, `.venv`/, `*.pyc`, etc.).

If deleted: Redundant with the root `.gitignore` but removes a safety net.

`backend/conftest.py`

REQUIRED

Purpose: Pytest fixtures/config loaded automatically by every test file.

If deleted: Backend test suite stops running.

`backend/pytest.ini`

REQUIRED

Purpose: Pytest configuration — declares `DJANGO_SETTINGS_MODULE=zarailink.settings` so tests can access the ORM.

If deleted: Tests fail on import because Django is not configured.

`backend/requirements.txt`

REQUIRED

Purpose: Pinned Python dependencies: Django 4.2, DRF, psycpg2, sentence-transformers, setfit, keybert, gliner, rapidfuzz, lightgbm, and others.

If deleted: Fresh installs cannot reproduce the dependency graph.

`backend/Dockerfile`

OPTIONAL

Purpose: Container image definition for the backend service.

If deleted: Cannot build a backend Docker image.

`backend/docker-compose.yml`

OPTIONAL

Purpose: Local `docker compose` file for standing up the backend container alongside Postgres.

If deleted: Breaks the compose-based dev workflow only.

`backend/entrypoint.sh`

OPTIONAL

Purpose: Shell script used as the Docker container entrypoint (waits for Postgres, runs migrations, launches Gunicorn).

If deleted: Breaks containerized deployment only.

`backend/package.json`

OPTIONAL

Purpose: Node manifest used solely by `generate_comparison_pdf.js` (Puppeteer for a PDF export). Not used by the Python backend itself.

If deleted: The Puppeteer-based comparison-PDF export fails.

`backend/package-lock.json`

OPTIONAL

Purpose: NPM lockfile for the above.

If deleted: Same.

4.2 Data loaders and seed fixtures

backend/populate_company_products.py**OPTIONAL****Purpose:** One-shot ORM script that fills `CompanyProduct` rows from transaction data. Used only after fresh seeding.*If deleted:* Cannot rebuild the `CompanyProduct` table from scratch.**backend/subscriptions_only.json****OPTIONAL****Purpose:** 17 KB partial fixture containing only subscription rows. Used for selective re-seeding.*If deleted:* Lose the ability to re-seed just the subscriptions table.

4.3 GraphML files

All four `.graphml` files at the backend root are read by live code in `trade_ledger` (link prediction, GNN embeddings). The three dead files (`company_company_graph`, `company_country_graph`, `product_country_graph`) were removed in the April 2026 cleanup.

backend/buyer_seller_graph.graphml**REQUIRED****Purpose:** Buyer-seller edge graph. Loaded by `trade_ledger.services.link_prediction` for the link-prediction API.*If deleted:* Link-prediction endpoint fails until regenerated by `python manage.py build_gnn_graphs`.**backend/company_product_graph.graphml****REQUIRED****Purpose:** Company ↔ product bipartite graph, consumed by the GNN embedding command.*If deleted:* GNN embedding regeneration breaks.**backend/product_co_trade_graph.graphml****REQUIRED****Purpose:** Product co-occurrence graph used by GNN embeddings.*If deleted:* Same.**backend/seller_product_graph.graphml****REQUIRED****Purpose:** Seller ↔ product graph written by `build_gnn_graphs`.*If deleted:* Regenerated automatically by the same management command.

4.4 Binaries and cross-platform leftovers

backend/generate_comparison_pdf.js**OPTIONAL****Purpose:** Puppeteer-driven Node script that renders a comparison report to PDF. Used only if a Django view shells out to it. No Python import of this file exists.*If deleted:* The Puppeteer-based PDF comparison export breaks. Shell-out path is the only dependency.

4.5 backend/ dev scripts — all DELETED

backend/ dev scripts — all DELETED

All files in this list have been removed from the repository as part of the dev-cruft cleanup. The scripts that previously lived here were standalone dev-time helpers not imported by any Django app, management command, or test file -- all classified **DEAD**. Impact: zero.

4.6 Stale debug output (.txt / .json / .log) — all DELETED

backend/ stale outputs — all DELETED

All files in this list have been removed from the repository as part of the dev-cruft cleanup. These were stale output logs written by the debug scripts during iterative debugging. None were read by any live code -- all classified **DEAD**. Impact: zero.

4.7 Backend sub-directories listed as folders

backend/staticfiles/**OPTIONAL****Purpose:** Django `collectstatic` output, regenerated by running the command.*If deleted:* Production static serving breaks until you re-run `collectstatic`.

backend/scripts/**OPTIONAL**

Purpose: Directory containing the SetFit training files: `train_intent_model.py` (the training script) and `training_data.csv` (the 30-sample training dataset). All other debug scripts that previously lived here have been removed.

If deleted: Lose the SetFit retraining path.

backend/templates/admin/base_site.html**REQUIRED**

Purpose: Django admin base-template override — customizes the admin site header/branding.

If deleted: Admin branding reverts to the default Django look. The admin still works.

backend/static/admin/css/custom_admin.css**OPTIONAL**

Purpose: Custom admin CSS override.

If deleted: Admin reverts to default styling.

backend/static/css/custom_admin.css**OPTIONAL**

Purpose: Duplicate admin CSS at another static path.

If deleted: Same. Likely one of these is unused.

4.8 utils/ — Utility package

A small utility package providing AI and Redis helpers imported by live Django code.

backend/utils/__init__.py**REQUIRED**

Purpose: Package marker — makes utils importable as `from utils.ai_service import AIService`.

If deleted: Imports from `utils.*` raise `ModuleNotFoundError`.

backend/utils/ai_service.py**OPTIONAL**

Purpose: `AIService` class wrapping the OpenRouter / OpenAI API. Provides `smart_search()` (used by `companies.views.CompanyViewSet` for AI-assisted directory search) and `get_embedding()` (used by the now-deleted `generate_embeddings` management command). Requires `OPENROUTER_API_KEY` in `.env`.

If deleted: AI smart-search in the company directory falls back to the standard ORM filter. No crash — the view handles the absence gracefully.

5 Other Top-Level Directories

5.1 scripts/ — Tooling

scripts/md_to_pdf.py

OPTIONAL

Purpose: WeasyPrint-based script for rendering markdown documents into styled PDFs. Invoked manually.

If deleted: Lose the ability to regenerate PDF documents from markdown sources.

5.2 Resources/ — SRS and documentation assets

Reference images and PDFs used in the Software Requirements Specification and other academic documentation. All marked **OPTIONAL** — no runtime impact, but they are the figures embedded in the SRS.

Resources/1.pdf

OPTIONAL

Reference material for the SRS (532 KB). — *Figure missing from the SRS.*

Resources/1.png

OPTIONAL

SRS figure (133 KB). — *Figure missing.*

Resources/2.pdf

OPTIONAL

Reference (428 KB). — *Figure missing.*

Resources/2.png

OPTIONAL

SRS figure (88 KB). — *Figure missing.*

Resources/3.jpeg

OPTIONAL

SRS image (50 KB). — *Figure missing.*

Resources/3.pdf

OPTIONAL

Reference (362 KB). — *Figure missing.*

Resources/4.pdf

OPTIONAL

Reference (457 KB). — *Figure missing.*

Resources/4.png

OPTIONAL

SRS figure (59 KB). — *Figure missing.*

Resources/5.pdf

OPTIONAL

Reference (578 KB). — *Figure missing.*

Resources/5.png

OPTIONAL

SRS figure (59 KB). — *Figure missing.*

5.3 docs/ — Audit source

The docs/ directory contains the LaTeX source and compiled PDF for this audit document. Build artefacts (.aux, .log, .out, .toc) are auto-generated and safe to delete.

docs/file_audit.tex

OPTIONAL

Purpose: LaTeX source for this file-audit document. Edit here and recompile to update the PDF.

If deleted: Cannot regenerate docs/file_audit.pdf without rewriting from scratch.

`docs/file_audit.pdf`

OPTIONAL

Purpose: Compiled PDF of this audit. Regenerated by running `pdflatex file_audit.tex` twice in `docs/`.

If deleted: Lose the rendered audit; the source `.tex` is still intact.

5.4 .github/ — CI configuration

`.github/workflows/ci.yml`

OPTIONAL

Purpose: GitHub Actions workflow that runs backend `pytest` and frontend `npm test` on push/PR to main.

If deleted: Continuous integration stops on the main branch. Local development is unaffected.

6 Backend — Django Apps

Seven live Django apps under `backend/`, registered in `zarailink/settings.INSTALLED_APPS`: (`accounts`, `companies`, `market_intel`, `subscriptions`, `trade_data`, `trade_ledger`, `trade_lens`, plus `search` which has its own section). The empty scaffold apps (`home`, `trade_directory`) have been removed as part of the April 2026 cleanup.

6.1 Django project package (`backend/zarailink/`)

The four-file nerve center of the project. None of these are optional.

backend/zarailink/__init__.py

REQUIRED

Purpose: Empty package marker for the Django project.

If deleted: Project stops being a Python package; Django refuses to boot.

backend/zarailink/settings.py

REQUIRED

Purpose: Full Django settings: `INSTALLED_APPS`, DRF, PostgreSQL creds, Redis auto-detection, CORS/CSRF, email, CKEditor 5, logging config, and the startup env-var noise suppression for TensorFlow/transformers/matplotlib. Hardcodes `DEBUG=True` for dev.

If deleted: Django has no configuration and refuses to start.

backend/zarailink/urls.py

REQUIRED

Purpose: Root URLConf — mounts `accounts`, `subscriptions`, `market-intel`, `companies`, `trade_ledger`, `trade-lens`, `search`, `ckeditor5`, the DRF login view, and a SPA catch-all that serves the React `index.html`.

If deleted: All HTTP routes 404.

backend/zarailink/wsgi.py

REQUIRED

Purpose: WSGI application factory for Gunicorn/`mod_wsgi`.

If deleted: Production servers cannot boot the Django app.

backend/zarailink/asgi.py

REQUIRED

Purpose: ASGI application factory for Daphne/Uvicorn.

If deleted: Async/ASGI servers cannot boot the app.

6.2 `accounts/` — Authentication & user management

Custom `User` model (email-based, with verification token and `token_balance` economy), `signup/login/logout/verify/resend/forgot` password REST endpoints using session auth, and an audit-log scaffold.

backend/accounts/__init__.py

REQUIRED

Purpose: Package marker.

If deleted: Django app loader cannot import the app.

backend/accounts/apps.py

REQUIRED

Purpose: `AppConfig` registering `accounts` with Django.

If deleted: `INSTALLED_APPS` registration fails.

backend/accounts/admin.py

REQUIRED

Purpose: Registers `User` and `UserAlertPreference` in the Django admin with fieldsets, bulk token-grant actions, and a `verify-users` action.

If deleted: Admin panel loses user management; APIs still work.

backend/accounts/models.py

REQUIRED

Purpose: Custom `User` model with email PK, `verification_token`, `token_balance` methods, plus `UserAlertPreference`. Referenced by `AUTH_USER_MODEL`.

If deleted: Entire app cannot start — Django crashes at import.

backend/accounts/forms.py

REQUIRED

Purpose: `UserRegisterForm` (extends Django's `UserCreationForm`), splits `name` into first/last and sets `is_active=False` on creation.

If deleted: `api_signup` view crashes on import.

backend/accounts/views.py

REQUIRED

Purpose: Session-auth API views: `api_signup`, `api_login`, `api_logout`, `api_check_auth`, `api_verify_email`, `api_forgot_password`,

api_resend_verification. Sends HTML verification emails.

If deleted: All `/accounts/*` endpoints 404; login, signup, email verification broken.

backend/accounts/urls.py

REQUIRED

Purpose: URL map for the accounts endpoints and the Django `password_reset_confirm/complete` views.

If deleted: Accounts URL include breaks; 404 on all auth routes.

backend/accounts/templates/registration/password_reset_complete.html

REQUIRED

Purpose: Success page for Django's built-in `password_reset_complete` view.

If deleted: Password reset completion page 500s.

backend/accounts/templates/registration/password_reset_confirm.html

REQUIRED

Purpose: Form template for Django's `password_reset_confirm` view.

If deleted: Password reset link flow 500s.

backend/accounts/tests/__init__.py

OPTIONAL

Purpose: Test package marker.

If deleted: Pytest collection of accounts tests breaks.

backend/accounts/tests/test_models.py

OPTIONAL

Purpose: Unit tests for User creation, verification token, token balance, and `UserAlertPreference`.

If deleted: Lose unit-test coverage.

backend/accounts/tests/test_views.py

OPTIONAL

Purpose: Integration tests for the signup/login/verify/forgot-password/resend flows.

If deleted: Lose integration-test coverage.

backend/accounts/migrations/__init__.py

REQUIRED

Purpose: Marks migrations/ as a package.

If deleted: Django migration loader cannot find accounts migrations.

backend/accounts/migrations/0001_initial.py

REQUIRED

Purpose: Schema: `CreateModel User` (email unique, `verification_token` UUID, etc.).

If deleted: Fresh databases cannot create the User table.

backend/accounts/migrations/0002_user_job_title_user_phone_number_useralertpreference.py

REQUIRED

Purpose: Adds `phone_number` and `job_title` to User; creates `UserAlertPreference`.

If deleted: Migration chain breaks.

backend/accounts/migrations/0003_user_token_balance.py

REQUIRED

Purpose: Adds `token_balance` (int, default 0) to User.

If deleted: Token economy column missing; migration chain breaks.

backend/accounts/migrations/0004_alter_user_managers.py

REQUIRED

Purpose: Resets the User model's `managers` list.

If deleted: Migration chain breaks.

6.3 companies/ — Trade directory

Company/directory models, DRF viewsets for browsing and filtering, AI-assisted search, token-gated key-contact unlocks, and several management commands — some live, some one-shot data patches.

backend/companies/__init__.py

REQUIRED

Purpose: Package marker.

If deleted: App cannot be imported.

backend/companies/apps.py

REQUIRED

Purpose: `CompaniesConfig`.

If deleted: App registration fails.

backend/companies/admin.py **REQUIRED**
Purpose: Registers Sector, CompanyRole, CompanyType, Company (with CKEditor 5), CompanyProduct, KeyContact (with unlock inline), KeyContactUnlock, Image.
If deleted: Admin loses directory management; APIs still work.

backend/companies/models.py **REQUIRED**
Purpose: Core directory models: Sector, CompanyRole, CompanyType, Image (polymorphic), Company (verification_status, sentiment, FKs), CompanyProduct (hsn_code), KeyContact, KeyContactUnlock, CompanyMetricsCache, IngestionLog, AdminAnnouncement. Registers a few models with `django-auditlog`.
If deleted: Trade Directory app cannot load; other apps that FK to Company also break.

backend/companies/serializers.py **REQUIRED**
Purpose: DRF serializers for Sector/Role/Type, Company list/detail, and a dynamic `KeyContactSerializer` that masks phone/email/whatsapp unless unlocked.
If deleted: Company and KeyContact API responses break.

backend/companies/views.py **REQUIRED**
Purpose: `CompanyViewSet` (filtered by country/sector/role/type, with optional AI smart-search), `KeyContactViewSet` with the token-deducting unlock action, Sector/Type/Role list views.
If deleted: `/api/companies/*`, `/api/key-contacts/*`, and the sector/role/type endpoints all 404.

backend/companies/urls.py **REQUIRED**
Purpose: `DefaultRouter` registration for the viewsets, plus lookup endpoints.
If deleted: URL include breaks; all directory endpoints 404.

backend/companies/tests/__init__.py **OPTIONAL**
Purpose: Test package marker.
If deleted: Pytest collection breaks for companies tests.

backend/companies/tests/test_views.py **OPTIONAL**
Purpose: Integration tests for company filtering, pagination, and key-contact unlock (with/without tokens).
If deleted: Lose test coverage.

backend/companies/management/__init__.py **REQUIRED**
Purpose: Package marker.
If deleted: `manage.py` cannot find companies commands.

backend/companies/management/commands/__init__.py **REQUIRED**
Purpose: Package marker.
If deleted: Commands undiscoverable.

backend/companies/management/commands/seed_roles.py **OPTIONAL**
Purpose: Seeds CompanyRole with the 5 canonical roles (Buyer, Supplier, Distributor, Manufacturer, Service Provider).
If deleted: Fresh-DB seed missing; can be reproduced by hand.

backend/companies/management/commands/setup_company_roles.py **OPTIONAL**
Purpose: Alternative seeder creating Buyers/Suppliers/Both and backfilling NULL roles. Redundant with `seed_roles.py`.
If deleted: None (duplicate).

backend/companies/migrations/__init__.py **REQUIRED**
Purpose: Package marker.
If deleted: Migration loader breaks.

backend/companies/migrations/0001_initial.py **REQUIRED**
Purpose: Schema: creates Sector, CompanyRole, CompanyType, Image, Company, CompanyProduct, KeyContact, KeyContactUnlock, CompanyMetricsCache, IngestionLog, AdminAnnouncement; initial indexes.
If deleted: Fresh DB cannot create companies schema.

backend/companies/migrations/0002_companyproduct_hsn_code.py **REQUIRED**
Purpose: Adds `CompanyProduct.hsn_code` (max_length 20).
If deleted: Migration chain breaks.

backend/companies/migrations/0003_company_market_sentiment.py

REQUIRED

Purpose: Adds `Company.market_sentiment` (default “Neutral”).

If deleted: Migration chain breaks.

6.4 market_intel/ — News & recommendations

Alert/NewsArticle/SavedAnalysis models (no endpoints or admin) plus a `UserInteraction` model and a single recommendations endpoint backed by Redis vector similarity over the logged interactions.

backend/market_intel/__init__.py

REQUIRED

Purpose: Package marker.

If deleted: App cannot import.

backend/market_intel/apps.py

REQUIRED

Purpose: `MarketIntelConfig`.

If deleted: App registration breaks.

backend/market_intel/models.py

REQUIRED

Purpose: `Alert` (severity, FK to `Sector`), `NewsArticle` (FK `Alert`), `SavedAnalysis` (user bookmarks), `UserInteraction` (user-company-action tracking).

If deleted: The interaction write in `companies.views` and the recommendations API both fail.

backend/market_intel/views.py

REQUIRED

Purpose: `recommendations_api` — averages Redis vectors from a user’s recent interactions and returns the top-k similar companies.

If deleted: `/api/market-intel/recommendations/` 404s.

backend/market_intel/urls.py

REQUIRED

Purpose: URLconf mapping `recommendations/` to the view.

If deleted: URL include breaks.

backend/market_intel/migrations/__init__.py

REQUIRED

Purpose: Package marker.

If deleted: Migration loader breaks.

backend/market_intel/migrations/0001_initial.py

REQUIRED

Purpose: Creates `Alert`, `NewsArticle`, `SavedAnalysis`.

If deleted: Fresh DB schema creation fails.

backend/market_intel/migrations/0002_userinteraction.py

REQUIRED

Purpose: Creates `UserInteraction`.

If deleted: Interaction table missing.

6.5 subscriptions/ — Plans, tokens, redeem codes

Subscription plans, user subscriptions, token purchases, and redeem codes that replace real payments for the MVP. Two APIs: list plans, redeem code.

backend/subscriptions/__init__.py

REQUIRED

Purpose: Package marker.

If deleted: App cannot import.

backend/subscriptions/apps.py

REQUIRED

Purpose: `SubscriptionsConfig`.

If deleted: App registration breaks.

backend/subscriptions/admin.py

REQUIRED

Purpose: Admin for `SubscriptionPlan`, `RedeemCode` (with batch code generation), `UserSubscription`, `TokenPurchase`.

If deleted: Admin loses subscription tooling; APIs still work.

backend/subscriptions/models.py

REQUIRED

Purpose: Plans, user subscriptions, token purchases, redeem codes (with `generate_code` and atomic `redeem()` that deposits tokens).

If deleted: Redeem/plan APIs and admin break.

backend/subscriptions/views.py Purpose: <code>list_plans</code> (public), <code>redeem_code</code> (authenticated, atomic), <code>generate_codes_view</code> (staff-only). <i>If deleted:</i> <code>/api/subscriptions/*</code> endpoints break.	REQUIRED
backend/subscriptions/urls.py Purpose: URL map for plans/redeem/generate-codes. <i>If deleted:</i> URL include breaks.	REQUIRED
backend/subscriptions/tests/__init__.py Purpose: Test marker. <i>If deleted:</i> Pytest collection breaks for this app.	OPTIONAL
backend/subscriptions/tests/test_views.py Purpose: Integration tests for listing plans and redeeming codes (valid/expired/already redeemed). <i>If deleted:</i> Lose test coverage.	OPTIONAL
backend/subscriptions/templatetags/__init__.py Purpose: Package marker for templatetags. <i>If deleted:</i> <code>admin_extras</code> tag unavailable.	REQUIRED
backend/subscriptions/templatetags/admin_extras.py Purpose: Template tag <code>get_subscription_plans</code> intended for admin templates. Not actually loaded anywhere in the current admin base template. <i>If deleted:</i> None.	OPTIONAL
backend/subscriptions/management/__init__.py Purpose: Package marker. <i>If deleted:</i> Commands undiscoverable.	REQUIRED
backend/subscriptions/management/commands/__init__.py Purpose: Package marker. <i>If deleted:</i> Commands undiscoverable.	REQUIRED
backend/subscriptions/management/commands/create_plans.py Purpose: Seeds the six canonical plans (500/5k/15k monthly, 6k/60k/900k annual) in PKR. <i>If deleted:</i> Fresh DB requires manual plan seeding.	OPTIONAL
backend/subscriptions/management/commands/generate_codes.py Purpose: CLI for bulk-generating redeem codes for a plan. <i>If deleted:</i> Admin bulk action is still available.	OPTIONAL
backend/subscriptions/migrations/__init__.py Purpose: Package marker. <i>If deleted:</i> Migration loader breaks.	REQUIRED
backend/subscriptions/migrations/0001_initial.py Purpose: Creates SubscriptionPlan, UserSubscription, TokenPurchase. <i>If deleted:</i> Fresh DB schema creation fails.	REQUIRED
backend/subscriptions/migrations/0002_redeemcode.py Purpose: Creates RedeemCode with indexed code + status. <i>If deleted:</i> Migration chain breaks.	REQUIRED

6.6 trade_data/ — Transactions & embeddings

The canonical transaction dataset — Product/Category/SubCategory/Item hierarchy, the `Transaction` table, and `CompanyEmbedding/ProductEmbedding` for ML. The app has no views or URLs — it is a pure model layer consumed by `trade_ledger`, `search`, and `market_intel`.

backend/trade_data/__init__.py

REQUIRED

Purpose: Package marker.

If deleted: App cannot import.

backend/trade_data/apps.py

REQUIRED

Purpose: TradeDataConfig.

If deleted: App registration breaks.

backend/trade_data/models.py

REQUIRED

Purpose: Legacy `HsToProductMap`, the Product/Category/SubCategory/Item hierarchy, `Transaction` (`trade_type`, `buyer`, `seller`, `origin/destination_country`, `qty_kg/qty_mt`, `usd/pkr`, composite indexes), `CompanyEmbedding`, `ProductEmbedding`.

If deleted: `trade_ledger`, `search`, and the recommendations endpoint all crash.

backend/trade_data/management/__init__.py

REQUIRED

Purpose: Package marker.

If deleted: Commands undiscoverable.

backend/trade_data/management/commands/__init__.py

REQUIRED

Purpose: Package marker.

If deleted: Commands undiscoverable.

backend/trade_data/migrations/__init__.py

REQUIRED

Purpose: Package marker.

If deleted: Migration loader breaks.

backend/trade_data/migrations/0001_initial.py

REQUIRED

Purpose: Schema: initial `HsToProductMap`, `Transaction`, legacy `AggProductMonthCountry`, `AggCompanyMonthProduct`.

If deleted: Fresh DB creation fails.

backend/trade_data/migrations/0002_remove_transaction_...

REQUIRED

Purpose: Heavy refactor of `Transaction` — removes old indexes/fields, adds `hs_code`, `buyer`, `seller`, `country` columns, `qty_kg`, `qty_mt`, `usd`; re-adds indexes.

If deleted: Migration chain breaks.

backend/trade_data/migrations/0003_alter_transaction_buyer_...

REQUIRED

Purpose: Widens `buyer`, `seller`, `country` `max_length`.

If deleted: Migration chain breaks.

backend/trade_data/migrations/0004_alter_transaction_category_...

REQUIRED

Purpose: `AlterField` on 5 `Transaction` text fields.

If deleted: Migration chain breaks.

backend/trade_data/migrations/0005_remove_hstoproductmap_product_...

REQUIRED

Purpose: Removes `HsToProductMap.product`.

If deleted: Migration chain breaks.

backend/trade_data/migrations/0006_product_productcategory_...

REQUIRED

Purpose: Creates `Product`, `ProductCategory`; removes `Transaction.category/hs_code/product`; creates `ProductSubCategory`, `ProductItem`; adds `Transaction.product_item` FK.

If deleted: Product hierarchy missing; `trade_ledger+search` crash.

backend/trade_data/migrations/0007_alter_product_name_...

REQUIRED

Purpose: Widens names and `hs_codes` across the product hierarchy.

If deleted: Migration chain breaks.

backend/trade_data/migrations/0008_transaction_trade_type.py Purpose: Adds <code>trade_type</code> (IMPORT/EXPORT, indexed) to <code>Transaction</code> . <i>If deleted:</i> Import/export split broken.	REQUIRED
backend/trade_data/migrations/0009_companyembedding_productembedding.py Purpose: Creates <code>CompanyEmbedding</code> and <code>ProductEmbedding</code> (JSONField embeddings). <i>If deleted:</i> GNN/similar-companies storage missing.	REQUIRED
backend/trade_data/migrations/0010_remove_aggcompanymonthproduct... Purpose: Removes legacy aggregate tables, adds <code>origin_country/destination_country</code> , fresh <code>Transaction</code> indexes. <i>If deleted:</i> Migration chain breaks.	REQUIRED
backend/trade_data/migrations/0011_alter_productsubcategory_hs_code.py Purpose: Widens <code>ProductSubCategory.hs_code</code> . <i>If deleted:</i> Migration chain breaks.	REQUIRED
backend/trade_data/migrations/0012_auto_20260317_2359.py Purpose: Enables Postgres <code>TrigramExtension</code> and <code>BtreeGinExtension</code> for fuzzy search. <i>If deleted:</i> Trigram/GIN indexes unavailable; search degrades.	REQUIRED
backend/trade_data/migrations/0013_transaction_tx_type_product_idx_and_more.py Purpose: Adds the three composite <code>Transaction</code> indexes (<code>trade_type, product_item</code>), (<code>product_item, trade_type, seller</code>), (<code>product_item, trade_type, buyer</code>) for hot search paths. <i>If deleted:</i> Search aggregation queries slow down.	REQUIRED

6.7 trade_ledger/ — Trade intelligence APIs

Exposes the Trade Intelligence endpoints: Explorer, Company Overview / Products / Partners / Trends, Compare Companies, Similar Companies, and Link Prediction. All views read live from `trade_data.Transaction` through service modules. The app has *no models.py* and *no actual migrations*. Several legacy files reference a nonexistent `trade_ledger.models` module — those are dead.

backend/trade_ledger/__init__.py Purpose: Package marker. <i>If deleted:</i> App cannot import.	REQUIRED
backend/trade_ledger/apps.py Purpose: <code>TradeLedgerConfig</code> . <i>If deleted:</i> App registration breaks.	REQUIRED
backend/trade_ledger/urls.py Purpose: URL map for <code>explorer/</code> , <code>company/<name>/overview</code> , <code>products</code> , <code>partners</code> , <code>trends</code> , <code>similar</code> , <code>potential-partners</code> , <code>network-influence</code> , <code>compare/</code> , <code>export-comparison-pdf/</code> , <code>product-clusters/</code> , <code>predict/sellers</code> , <code>buyers</code> , <code>methods</code> . <i>If deleted:</i> All <code>/api/explorer/</code> and <code>/api/company/*</code> endpoints 404.	REQUIRED
backend/trade_ledger/views.py Purpose: Function views wrapping the service modules — the entire Trade Intelligence API surface. Also decorates responses with <code>CompanyEmbedding</code> cluster/pagerank data. <i>If deleted:</i> All trade-intelligence endpoints 404/500.	REQUIRED
backend/trade_ledger/pdf_export_view.py Purpose: <code>export_comparison_pdf</code> — spawns a Node subprocess invoking <code>backend/generate_comparison_pdf.js</code> (Puppeteer) to render a company-comparison PDF and streams it back. <i>If deleted:</i> <code>/api/export-comparison-pdf/</code> breaks.	REQUIRED
backend/trade_ledger/services/__init__.py Purpose: Package marker. <i>If deleted:</i> Services imports break.	REQUIRED
backend/trade_ledger/services/company.py Purpose: Company-overview aggregations (volume, revenue, MoM growth, top products and partners, partner geography, monthly trend).	REQUIRED

If deleted: Company overview API breaks.

backend/trade_ledger/services/compare.py

REQUIRED

Purpose: Per-company comparison aggregation (volume, value, partner diversity HHI, MoM growth, price stddev, pagerank).

If deleted: Compare API breaks.

backend/trade_ledger/services/explorer.py

REQUIRED

Purpose: `get_explorer_companies` — paginated company listing with direction/date/country/product filters and buyer/seller role classification.

If deleted: Explorer API breaks.

backend/trade_ledger/services/filters.py

REQUIRED

Purpose: Single place that applies direction/company/date/country/product filters onto a Transaction queryset.

If deleted: All trade_ledger services fail to filter.

backend/trade_ledger/services/gnn.py

REQUIRED

Purpose: `get_company_embedding` and `get_similar_companies` — cosine similarity over CompanyEmbedding rows with fuzzy-name matching.

If deleted: Similar-companies and potential-partners APIs break.

backend/trade_ledger/services/link_prediction.py

REQUIRED

Purpose: Node2Vec / common-neighbors / product-cotrade / Jaccard / preferential-attachment / combined buyer-seller predictors using NetworkX GraphML + CompanyEmbedding.

If deleted: `/api/predict/*` APIs break.

backend/trade_ledger/services/partners.py

REQUIRED

Purpose: YoY growth, top partners, partner geography, product mix per partner.

If deleted: Company-partners API breaks.

backend/trade_ledger/services/products.py

REQUIRED

Purpose: Product-level analytics: YoY growth, per-company product performance, avg-price trends, volume share, co-traded products, product clusters, partner matrix.

If deleted: Company-products API breaks.

backend/trade_ledger/services/trends.py

REQUIRED

Purpose: `get_volume_price_monthly`, `get_yoy_growth_by_quarter` feeding the trends endpoint.

If deleted: Company-trends API breaks.

backend/trade_ledger/tests/__init__.py

OPTIONAL

Purpose: Test marker.

If deleted: Pytest collection breaks for this app.

backend/trade_ledger/tests/test_views.py

OPTIONAL

Purpose: Integration tests for Explorer / company / compare / GNN / link-prediction endpoints.

If deleted: Lose test coverage.

backend/trade_ledger/tests/test_ai_accuracy.py

OPTIONAL

Purpose: Pytest ML-accuracy tests (confidence bounds, non-random predictions, edge cases).

If deleted: Lose test coverage.

backend/trade_ledger/tests/test_link_prediction.py

OPTIONAL

Purpose: Per-method unit tests for link prediction.

If deleted: Lose test coverage.

backend/trade_ledger/management/__init__.py

REQUIRED

Purpose: Package marker.

If deleted: Commands undiscoverable.

backend/trade_ledger/management/commands/__init__.py

REQUIRED

Purpose: Package marker.

If deleted: Commands undiscoverable.

backend/trade_ledger/management/commands/build_gnn_graphs.py**OPTIONAL**

Purpose: Builds company-product and buyer-seller GraphML files from Transaction rows using NetworkX. Required only to regenerate the `.graphml` assets.

If deleted: Regeneration of link-prediction graphs breaks.

backend/trade_ledger/management/commands/generate_gnn_embeddings.py**OPTIONAL**

Purpose: Reads GraphML, runs Node2Vec and HDBSCAN, writes CompanyEmbedding/ProductEmbedding rows. Supports a `-fast` mode.

If deleted: GNN prep pipeline missing; app still serves the precomputed rows.

backend/trade_ledger/migrations/__init__.py**REQUIRED**

Purpose: Package marker — no actual migration files because the app has no models.

If deleted: Django startup migration check fails.

6.8 trade_lens/ — Product analytics

Isolated TradeLensProduct and TradeLensTransaction tables (independent from `trade_data.Transaction`) that feed the Trade Lens dashboard widgets. Single ReadOnly ViewSet at `/api/trade-lens/products/` with multiple `@action` endpoints for summary, overview, comparison, global view, details.

backend/trade_lens/__init__.py**REQUIRED**

Purpose: Package marker.

If deleted: App cannot import.

backend/trade_lens/apps.py**REQUIRED**

Purpose: TradeLensConfig.

If deleted: App registration breaks.

backend/trade_lens/admin.py**REQUIRED**

Purpose: Admin for TradeLensProduct and TradeLensTransaction.

If deleted: Admin loses TradeLens data management; APIs still work.

backend/trade_lens/models.py**REQUIRED**

Purpose: TradeLensProduct (name, hs_code, category) and TradeLensTransaction (trade_date, price, quantity_mt, auto-computed total_value_usd, buyer/seller/country/port/province, indexed).

If deleted: App breaks entirely.

backend/trade_lens/serializers.py**REQUIRED**

Purpose: Product and Transaction serializers plus flat Overview/Summary/Comparison/GlobalView serializers.

If deleted: TradeLens API responses break.

backend/trade_lens/views.py**REQUIRED**

Purpose: TradeLensProductViewSet with pagination, annotations for import/export volume/value/count, and `@action` endpoints for the dashboard widgets.

If deleted: `/api/trade-lens/*` endpoints break.

backend/trade_lens/urls.py**REQUIRED**

Purpose: DefaultRouter registration.

If deleted: URL include breaks.

backend/trade_lens/management/__init__.py**REQUIRED**

Purpose: Package marker.

If deleted: Commands undiscoverable.

backend/trade_lens/management/commands/__init__.py**REQUIRED**

Purpose: Package marker.

If deleted: Commands undiscoverable.

backend/trade_lens/migrations/__init__.py**REQUIRED**

Purpose: Package marker.

If deleted: Migration loader breaks.

backend/trade_lens/migrations/0001_initial.py

REQUIRED

Purpose: Creates TradeLensProduct and TradeLensTransaction with six indexes.

If deleted: TradeLens tables cannot be created.

7 Frontend (frontend/)

Create-React-App project (React 19 + Tailwind + Recharts + react-simple-maps + jspdf + framer-motion + react-router v7). Live entry point is `src/index.js` → `src/App.js` → route components. All API calls target the backend via `REACT_APP_API_BASE_URL`.

7.1 Frontend root (frontend/)

frontend/package.json Purpose: CRA manifest declaring dependencies and npm scripts. <i>If deleted:</i> App cannot install, build, or run.	REQUIRED
frontend/package-lock.json Purpose: Locked dependency tree pinned for deterministic installs. <i>If deleted:</i> Non-reproducible installs; not strictly fatal but strongly recommended.	REQUIRED
frontend/.env Purpose: Default dev env providing <code>REACT_APP_API_BASE_URL=http://localhost:8000</code> . <i>If deleted:</i> API calls resolve to <code>undefined/...</code> and fail.	REQUIRED
frontend/.env.local Purpose: Local override mirror of <code>.env</code> — CRA gives it higher precedence. <i>If deleted:</i> Harmless; <code>.env</code> still supplies the same value.	OPTIONAL
frontend/.env.production Purpose: Production env pointing <code>REACT_APP_API_BASE_URL</code> at the ngrok tunnel used by the deployed build. <i>If deleted:</i> Production build hits <code>localhost:8000</code> and is broken in deployed environments.	REQUIRED
frontend/.gitignore Purpose: Excludes <code>node_modules</code> , <code>build</code> , <code>coverage</code> , <code>.env.local</code> , and log files from git. <i>If deleted:</i> Risk of committing build artifacts/secrets; no runtime impact.	OPTIONAL
frontend/Dockerfile Purpose: Container spec (<code>node:18-alpine</code> , installs deps, runs <code>npm start</code> on 3000). <i>If deleted:</i> Docker workflow breaks; <code>npm start</code> still works locally.	OPTIONAL
frontend/postcss.config.js Purpose: PostCSS pipeline wiring Tailwind and autoprefixer. <i>If deleted:</i> Tailwind directives stop compiling; all styling breaks.	REQUIRED
frontend/tailwind.config.js Purpose: Tailwind config scanning <code>src/**/*.js,jsx,ts,tsx</code> . <i>If deleted:</i> Tailwind utility classes get purged; the UI loses all utility styling.	REQUIRED

7.2 frontend/public/

frontend/public/index.html Purpose: CRA HTML template with <code>#root</code> mount, Google Fonts preconnect, favicon + manifest links. <i>If deleted:</i> CRA build fails — no HTML shell for the React app.	REQUIRED
frontend/public/favicon.ico Purpose: Browser tab icon. <i>If deleted:</i> Missing favicon 404; app still runs.	OPTIONAL
frontend/public/logo192.png Purpose: 192x192 PWA/apple-touch icon. <i>If deleted:</i> Manifest icon 404; app still runs.	OPTIONAL

frontend/public/logo512.png Purpose: 512x512 PWA icon. <i>If deleted:</i> Manifest icon 404.	OPTIONAL
frontend/public/manifest.json Purpose: PWA web-app manifest (stock CRA default). <i>If deleted:</i> PWA install prompt disabled.	OPTIONAL
frontend/public/robots.txt Purpose: Crawler directives (allow all). <i>If deleted:</i> None.	OPTIONAL

7.3 frontend/src/ — root files

frontend/src/index.js Purpose: React entry point. Creates the root, wraps App in StrictMode, imports index.css and reportWebVitals. <i>If deleted:</i> No mount point; CRA build fails.	REQUIRED
frontend/src/index.css Purpose: Tailwind directives plus the Zarailink design-system CSS variables for light/dark themes and global body/scrollbar styles. <i>If deleted:</i> Tailwind utilities and all theme tokens disappear; the site is unstyled.	REQUIRED
frontend/src/App.js Purpose: Root component defining react-router routes, Protected/Public guards, Suspense wrapper, and the Theme/Toast/Auth providers. <i>If deleted:</i> No routing; nothing renders.	REQUIRED
frontend/src/App.css Purpose: Default CRA stylesheet (spinning-logo animation). Not imported anywhere. <i>If deleted:</i> None.	DEAD
frontend/src/reportWebVitals.js Purpose: CRA web-vitals stub called from index.js with no callback. <i>If deleted:</i> Must also remove the call in index.js.	OPTIONAL
frontend/src/setupTests.js Purpose: Jest setup — imports jest-dom, mocks matchMedia, IntersectionObserver, ResizeObserver, scrollTo. <i>If deleted:</i> Component tests that rely on these browser APIs fail.	OPTIONAL

7.4 src/context/ — Global React contexts

frontend/src/context/AuthContext.js Purpose: AuthProvider and useAuth — session auth via credentialed fetch to /accounts/api/check-auth, /login, /logout. Exposes user, tokenBalance, login, logout, isAuthenticated, refreshUser. <i>If deleted:</i> App.js import fails; entire app fails to mount.	REQUIRED
frontend/src/context/ThemeContext.js Purpose: ThemeProvider and useTheme — persists dark-mode preference to localStorage and toggles :root.dark. <i>If deleted:</i> App.js import fails; theming system breaks.	REQUIRED
frontend/src/context/__tests__/ThemeContext.test.js Purpose: Jest tests for an inline re-implementation of ThemeContext. <i>If deleted:</i> None.	OPTIONAL

7.5 src/hooks/ — Utility hooks

Local-storage-backed helpers for filters, watchlist, notes, activity history, offline detection, debounce, toasts, and onboarding tour. None of them call the backend.

frontend/src/hooks/index.js Purpose: Barrel re-export of all hooks. <i>If deleted:</i> Barrel consumers break; direct imports still work.	OPTIONAL
frontend/src/hooks/useDebounce.js Purpose: useDebounce/useDebounceCallback for typeahead inputs. <i>If deleted:</i> TradeLedger and search components throw import errors.	REQUIRED
frontend/src/hooks/useFilterPersistence.js Purpose: Persists filter objects to localStorage. <i>If deleted:</i> Filters reset on reload.	OPTIONAL
frontend/src/hooks/useToast.js Purpose: Thin wrapper around react-hot-toast. <i>If deleted:</i> Consumers must call toast() directly.	OPTIONAL
frontend/src/hooks/useWatchlist.js Purpose: Manages the zarailink-watchlist localStorage list. <i>If deleted:</i> Watchlist page and WatchlistButton flows break.	REQUIRED
frontend/src/hooks/useActivityHistory.js Purpose: Tracks the last 50 actions in zarailink-activity. <i>If deleted:</i> Activity-history features disappear.	OPTIONAL
frontend/src/hooks/useCompanyNotes.js Purpose: CRUD for per-company notes in zarailink-notes. Consumed by CompanyNotes. <i>If deleted:</i> CompanyNotes component crashes on import.	REQUIRED
frontend/src/hooks/useOnboardingTour.js Purpose: driver.js-based guided tour gated by a zarailink-onboarding-completed flag. <i>If deleted:</i> Onboarding tour disappears.	OPTIONAL
frontend/src/hooks/useOffline.js Purpose: Tracks navigator.onLine and caches data in zarailink-offline-data. <i>If deleted:</i> Offline detection removed.	OPTIONAL
frontend/src/hooks/__tests__/useDebounce.test.js Purpose: Jest tests for an inline useDebounce. <i>If deleted:</i> None.	OPTIONAL
frontend/src/hooks/__tests__/useToast.test.js Purpose: Jest tests for toast helper behavior. <i>If deleted:</i> None.	OPTIONAL
frontend/src/hooks/__tests__/useWatchlist.test.js Purpose: Jest tests for watchlist add/remove. <i>If deleted:</i> None.	OPTIONAL

7.6 src/services/ — API layer

frontend/src/services/api.js Purpose: Shared axios client with baseUrl = \${REACT_APP_API_BASE_URL}/api and withCredentials: true for session cookies. <i>If deleted:</i> searchService imports break; Search module fails.	REQUIRED
frontend/src/services/searchService.js Purpose: search(), getSupplierDetails(), compareSuppliers() — used by SearchResults, DealDetail, ComparePage. <i>If deleted:</i> All three search screens throw on import.	REQUIRED

7.7 src/utils/

frontend/src/utils/exportUtils.js**REQUIRED****Purpose:** convertToCSV, downloadCSV, downloadPDF used by the shared ExportButton.*If deleted:* ExportButton import fails; any screen importing it crashes.**frontend/src/utils/formatUtils.js****OPTIONAL****Purpose:** isNA / formatOrNull / displayValue helpers for rendering possibly-missing API values.*If deleted:* Consumers must inline these checks.

7.8 src/styles/

frontend/src/styles/animations.css**OPTIONAL****Purpose:** Global button-press and transition animation classes.*If deleted:* Subtle animations lost.

7.9 src/components/Auth/ — Authentication screens

frontend/src/components/Auth/Signup.js**REQUIRED****Purpose:** Signup form with country picker and POST to /accounts/api/signup/.*If deleted:* /signup route unusable.**frontend/src/components/Auth/Login.js****REQUIRED****Purpose:** Login form calling useAuth().login(); handles email-not-verified redirect.*If deleted:* Login flow broken.**frontend/src/components/Auth/ForgotPassword.js****REQUIRED****Purpose:** Email-entry form requesting a reset link.*If deleted:* /forgot-password route fails.**frontend/src/components/Auth/ResetPassword.js****REQUIRED****Purpose:** Token-based reset form hit from /reset-password/:token.*If deleted:* Reset-password route fails.**frontend/src/components/Auth/EmailVerification.js****REQUIRED****Purpose:** “Check your inbox” screen with resend button.*If deleted:* /email-verification route fails.**frontend/src/components/Auth/VerifyEmailSuccess.js****REQUIRED****Purpose:** Handles /verify-email/:token deep link; calls backend to confirm.*If deleted:* Email verification success route fails.

7.10 src/components/Common/ — Design system

Re-usable UI primitives: modal, breadcrumb, pagination, sort selector, skeleton loaders, empty states, charts, export/share/verification/watchlist buttons, and the react-hot-toast provider. Several files are 0-byte placeholders for planned but unbuilt widgets.

frontend/src/components/Common/index.js**OPTIONAL****Purpose:** Barrel re-export of all common primitives.*If deleted:* Barrel consumers break.**frontend/src/components/Common/Breadcrumb.js****REQUIRED****Purpose:** Location-aware breadcrumb mapping route segments to labels.*If deleted:* Multiple page imports crash.**frontend/src/components/Common/Breadcrumb.css****REQUIRED****Purpose:** Breadcrumb styles.*If deleted:* Renders unstyled.**frontend/src/components/Common/Charts.js****OPTIONAL****Purpose:** Recharts wrappers: TrendLineChart, ComparisonBarChart, DistributionPieChart, StatsCard.*If deleted:* Wrapper charts gone; most TI pages use Recharts directly.

frontend/src/components/Common/Charts.css Purpose: Wrapper chart styles. <i>If deleted:</i> None major.	OPTIONAL
frontend/src/components/Common/EmptyState.js Purpose: EmptyState + NoCompaniesFound/NoSearchResults/ErrorState variants. <i>If deleted:</i> Directory/Ledger/Watchlist/TradeLens screens crash.	REQUIRED
frontend/src/components/Common/EmptyState.css Purpose: Empty-state styles. <i>If deleted:</i> Renders unstyled.	REQUIRED
frontend/src/components/Common/ExportButton.js Purpose: CSV/PDF export dropdown backed by exportUtils. <i>If deleted:</i> Directory/Ledger/Watchlist fail to import.	REQUIRED
frontend/src/components/Common/ExportButton.css Purpose: Dropdown styles. <i>If deleted:</i> Renders unstyled.	REQUIRED
frontend/src/components/Common/Modal.js Purpose: Shared Modal + UnlockConfirmModal/SuccessModal/InsufficientTokensModal/ErrorModal used by CompanyProfile unlock flow and Subscription redeem flow. <i>If deleted:</i> CompanyProfile and Subscription pages crash.	REQUIRED
frontend/src/components/Common/Modal.css Purpose: Modal styles. <i>If deleted:</i> Modals unstyled.	REQUIRED
frontend/src/components/Common/Pagination.js Purpose: Reusable paginator with page-size selector. <i>If deleted:</i> FindBuyers/FindSuppliers/TradeLedger fail to import.	REQUIRED
frontend/src/components/Common/Pagination.css Purpose: Paginator styles. <i>If deleted:</i> Unstyled.	REQUIRED
frontend/src/components/Common/ShareButton.js Purpose: Copy-to-clipboard share button for a given URL. <i>If deleted:</i> Share action gone; rest of app works.	OPTIONAL
frontend/src/components/Common/ShareButton.css Purpose: Share button styles. <i>If deleted:</i> Minor styling loss.	OPTIONAL
frontend/src/components/Common/Skeleton.js Purpose: Skeleton/SkeletonText/SkeletonCard/SkeletonTable loading placeholders. <i>If deleted:</i> FindBuyers, FindSuppliers, TradeLens crash.	REQUIRED
frontend/src/components/Common/Skeleton.css Purpose: Shimmer animations. <i>If deleted:</i> Skeletons render as static boxes.	REQUIRED
frontend/src/components/Common/SortSelector.js Purpose: Labeled sort <select>. <i>If deleted:</i> FindBuyers/FindSuppliers/Watchlist crash.	REQUIRED
frontend/src/components/Common/SortSelector.css Purpose: Sort-selector styles. <i>If deleted:</i> Unstyled.	REQUIRED

frontend/src/components/Common/ToastProvider.js Purpose: Wraps children in a themed react-hot-toast Toaster. <i>If deleted:</i> App.js import fails; toasts disappear.	REQUIRED
frontend/src/components/Common/VerificationBadge.js Purpose: Verified/pending badge chip component. <i>If deleted:</i> Directory cards lose the badge.	OPTIONAL
frontend/src/components/Common/VerificationBadge.css Purpose: Badge styles. <i>If deleted:</i> Minor styling loss.	OPTIONAL
frontend/src/components/Common/WatchlistButton.js Purpose: Bookmark toggle used across directory, ledger, and intelligence pages. <i>If deleted:</i> Multiple screens fail to import.	REQUIRED
frontend/src/components/Common/WatchlistButton.css Purpose: Button styles. <i>If deleted:</i> Unstyled.	REQUIRED
frontend/src/components/Common/__tests__/EmptyState.test.js Purpose: Unit tests for EmptyState. <i>If deleted:</i> None.	OPTIONAL
frontend/src/components/Common/__tests__/Pagination.test.js Purpose: Unit tests for Pagination. <i>If deleted:</i> None.	OPTIONAL
frontend/src/components/Common/__tests__/VerificationBadge.test.js Purpose: Unit tests for VerificationBadge. <i>If deleted:</i> None.	OPTIONAL
frontend/src/components/Common/__tests__/WatchlistButton.test.js Purpose: Unit tests for WatchlistButton. <i>If deleted:</i> None.	OPTIONAL

7.11 src/components/Dashboard/ — Landing page

frontend/src/components/Dashboard/Dashboard.js Purpose: Authenticated home page with search autocomplete and feature-navigation cards. <i>If deleted:</i> /dashboard route fails.	REQUIRED
frontend/src/components/Dashboard/Dashboard.css Purpose: Dashboard styles, reused by the Search screens via relative import. <i>If deleted:</i> Dashboard, SearchResults, DealDetail, ComparePage render unstyled.	REQUIRED

7.12 src/components/Layout/ — App chrome

frontend/src/components/Layout/Navbar.js Purpose: Top navigation bar wiring AuthContext + ThemeContext + react-router, with dropdowns for Directory / Intelligence and framer-motion animated mobile menu. <i>If deleted:</i> Every authenticated page crashes.	REQUIRED
frontend/src/components/Layout/Navbar.css Purpose: Navbar styles. <i>If deleted:</i> Unstyled.	REQUIRED

7.13 src/components/Search/ — Search UI

frontend/src/components/Search/SearchHome.js**REQUIRED****Purpose:** Search landing page with product-suggest autocomplete and scope selector (WORLDWIDE/PAKISTAN).*If deleted:* /search route fails.**frontend/src/components/Search/SearchResults.js****REQUIRED****Purpose:** Paginated results consuming `searchService.search()` with filters, sort, and multi-select compare.*If deleted:* /search/results fails.**frontend/src/components/Search/DealDetail.js****REQUIRED****Purpose:** Supplier detail view with deal history, trends, and jsPDF-exported report (~52 KB — largest frontend file).*If deleted:* Supplier detail route fails.**frontend/src/components/Search/ComparePage.js****REQUIRED****Purpose:** Side-by-side comparison with Recharts trend lines and PDF export.*If deleted:* Compare route fails.

7.14 src/components/Subscriptions/

frontend/src/components/Subscriptions/Subscription.js**REQUIRED****Purpose:** Token/subscription page listing plans and driving the redeem-code modal.*If deleted:* /subscription route fails.**frontend/src/components/Subscriptions/Subscription.css****REQUIRED****Purpose:** Subscription page styles.*If deleted:* Unstyled.

7.15 src/components/TradeDirectory/

frontend/src/components/TradeDirectory/FindBuyers.js**REQUIRED****Purpose:** Paginated buyer directory with filters, sort, export, watchlist, breadcrumb.*If deleted:* /trade-directory/find-buyers fails.**frontend/src/components/TradeDirectory/FindBuyers.css****REQUIRED****Purpose:** FindBuyers styles.*If deleted:* Unstyled.**frontend/src/components/TradeDirectory/FindSuppliers.js****REQUIRED****Purpose:** Supplier directory mirror of FindBuyers.*If deleted:* FindSuppliers route fails.**frontend/src/components/TradeDirectory/FindSuppliers.css****REQUIRED****Purpose:** FindSuppliers styles.*If deleted:* Unstyled.**frontend/src/components/TradeDirectory/CompanyProfile.js****REQUIRED****Purpose:** Company detail with the key-contact unlock flow (four modal variants).*If deleted:* /trade-directory/company/:id fails.**frontend/src/components/TradeDirectory/CompanyProfile.css****REQUIRED****Purpose:** Company-profile styles.*If deleted:* Unstyled.**frontend/src/components/TradeDirectory/CompanyNotes.js****OPTIONAL****Purpose:** Inline note editor backed by `useCompanyNotes`.*If deleted:* CompanyProfile loses the notes widget.**frontend/src/components/TradeDirectory/CompanyNotes.css****OPTIONAL****Purpose:** Notes styles.*If deleted:* Minor styling loss.

7.16 src/components/TradeIntelligence/

The biggest module in the frontend — graph-driven analytics over the trade-transaction corpus. Company pages (Overview with `react-simple-maps`, Products/Partners with Recharts, Compare Companies with RadarChart), LinkPrediction (Node2Vec suggestions), TradeLedger (explorer), and the TradeLens suite (built on `react-chartjs-2`).

frontend/src/components/TradeIntelligence/TradeIntelligence.css

REQUIRED

Purpose: Shared stylesheet for most TI pages.

If deleted: Most Trade Intelligence pages render unstyled.

frontend/src/components/TradeIntelligence/TradeLedgerTable.css

OPTIONAL

Purpose: TradeLedger-only table styles.

If deleted: Table appearance degrades.

frontend/src/components/TradeIntelligence/CompanyOverview.js

REQUIRED

Purpose: Overview with world map (`react-simple-maps`) and summary stats.

If deleted: Company overview route fails.

frontend/src/components/TradeIntelligence/CompanyProducts.js

REQUIRED

Purpose: Product-mix Bar/Pie/Line charts (Recharts).

If deleted: Company products route fails.

frontend/src/components/TradeIntelligence/CompanyPartners.js

REQUIRED

Purpose: Partner analysis with Recharts LineChart.

If deleted: Company partners route fails.

frontend/src/components/TradeIntelligence/CompareCompanies.js

REQUIRED

Purpose: Two-company comparison (RadarChart + BarChart).

If deleted: Compare route fails.

frontend/src/components/TradeIntelligence/LinkPrediction.js

REQUIRED

Purpose: UI for the Node2Vec/combined link-prediction API.

If deleted: Link-prediction route fails.

frontend/src/components/TradeIntelligence/TradeLedger.js

REQUIRED

Purpose: Transaction ledger explorer (pagination, export, watchlist, debounced search).

If deleted: Ledger route fails.

frontend/src/components/TradeIntelligence/TradeLens.js

REQUIRED

Purpose: Product-analytics index listing tradable products with KPIs.

If deleted: TradeLens index route fails.

frontend/src/components/TradeIntelligence/TradeLensOverview.js

REQUIRED

Purpose: Chart.js line-chart overview for a product.

If deleted: Overview route fails.

frontend/src/components/TradeIntelligence/TradeLensSummary.js

REQUIRED

Purpose: Line + Bar summary view.

If deleted: Summary route fails.

frontend/src/components/TradeIntelligence/TradeLensComparison.js

REQUIRED

Purpose: Multi-select product comparison.

If deleted: Comparison route fails.

frontend/src/components/TradeIntelligence/TradeLensDetails.js

REQUIRED

Purpose: Detailed transactions + top buyers/sellers for a product.

If deleted: Details route fails.

frontend/src/components/TradeIntelligence/TradeLensGlobal.js

REQUIRED

Purpose: Global trade view (worldwide importers/exporters).

If deleted: Global route fails.

frontend/src/components/TradeIntelligence/TradePulse.js

OPTIONAL

Purpose: “Trade Pulse, coming soon...” placeholder page.

If deleted: /trade-intelligence/pulse route import fails — safe to delete only if you also remove the route.

7.17 src/components/Watchlist/

frontend/src/components/Watchlist/Watchlist.js

REQUIRED

Purpose: Watchlist page listing saved companies with sort, export, and clear actions.

If deleted: /watchlist route fails.

frontend/src/components/Watchlist/Watchlist.css

REQUIRED

Purpose: Watchlist styles.

If deleted: Unstyled.

8 Key Findings & Cleanup Recommendations

8.1 What's broken but still sitting on disk

All previously identified broken files have been removed as part of the April 2026 cleanup.

8.2 Quantified cleanup estimates

All items listed in this section have now been completed as part of the April 2026 cleanup. The following work was carried out:

- **~60 debug scripts** at `backend/*.py` — removed.
- **~60 stale output files** (`.txt`, `.json`, `.log`) at `backend/` — removed.
- **backend/ngrok.exe** (32 MB Windows binary) — removed.
- **3 dead .graphml files** (`company_company`, `company_country`, `product_country`) — removed.
- **backend/utils/redis_client.py** — removed (only caller was the deleted `index_companies.py`).
- **backend/search/models/setfit_intent_classifier/** (1.7 GB) — removed.
- **3 .graphml files** (`company_company_graph`, `company_country_graph`, `product_country_graph`) — removed.
- **0-byte frontend placeholder components**: all removed (`ProtectedRoute.js`, `ContactUnlockModal.js`, `TokenPurchaseModal.js`, `MarketBrief.js`, `Sidebar.js`, `KeyContactsTab.js`, `OverviewTab.js`, `SubscriptionPage.js`).
- **Broken backend files** with nonexistent model references — removed.
- **home/ and trade_directory/** empty scaffold apps — removed entirely.

8.3 What you must not delete

- Everything in `backend/zarailink/` — the Django project nerve center.
- All `migrations/*.py` files for the 7 live apps — the DB schema chain.
- `backend/.env` — sources `SECRET_KEY` for `settings.py`.
- `backend/search/models/intent_model/` — the trained SetFit model.
- `backend/manage.py`, `backend/conftest.py`, `backend/pytest.ini`, `backend/requirements.txt`.
- Frontend: everything in `src/context/`, `src/services/`, and all route components referenced by `App.js`.
- `frontend/.env` and `frontend/.env.production` — define `REACT_APP_API_BASE_URL`.
- `frontend/postcss.config.js` and `frontend/tailwind.config.js` — without them, all styling breaks.

8.4 Note on counts

This audit reflects the live source tree after the April 2026 cleanup. File counts exclude `.git/`, `node_modules/`, `__pycache__/`, `.venv/`, `frontend/build/`, and the `backend/search/models/intent_model/` bundle (already listed as a single entry). The trained SetFit model, package lock files, and other large assets are enumerated as single entries rather than per internal file.